Computer Networking

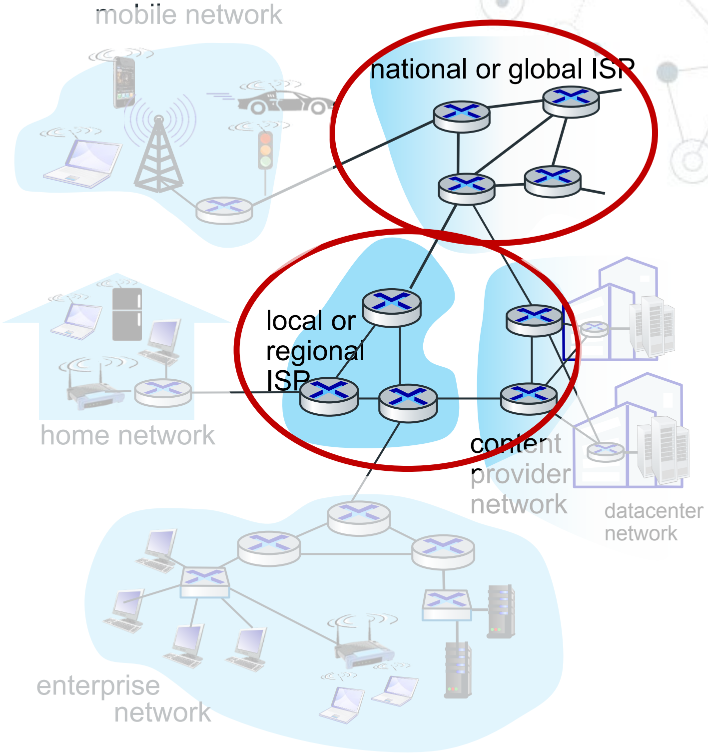
Chapter 1: Computer Networks and the Internet

Overview/roadmap:

- What *is* the Internet?
- What *is* a protocol?
- The Network edge: hosts, access network, physical media
- The Network core: packet/circuit switching, internet structure
- Performance: loss, delay, throughput
- Security
- Protocol layers, service models

The network core

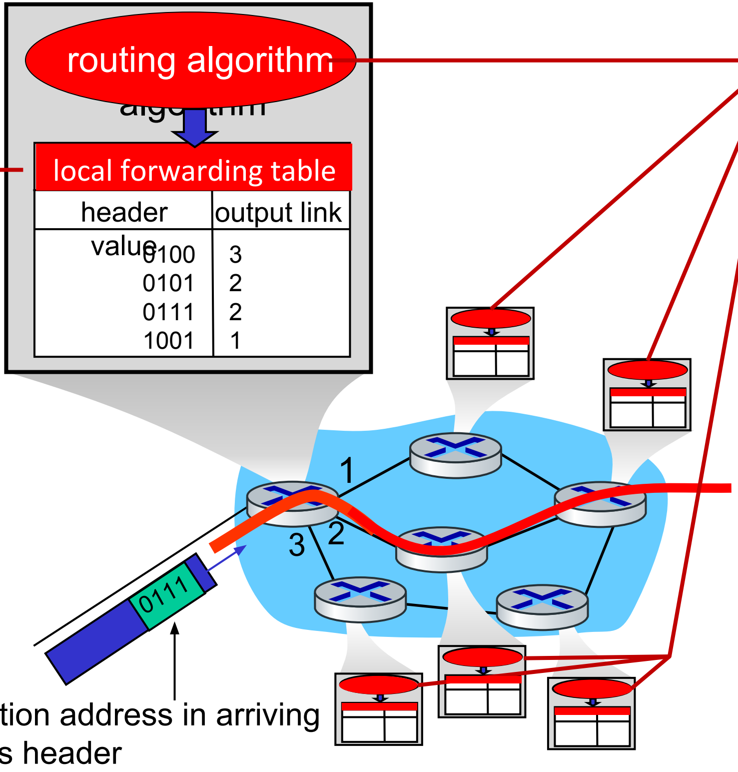
- mesh of interconnected routers
- packet-switching: hosts break application-layer messages into *packets*
  - forward packets from one router to the next, across links on path from source to destination
  - each packet transmitted at full link capacity



Two key network-core functions

Forwarding:

- *local* action: move arriving packets from router's input link to appropriate router output link



Routing:

- *global* action: determine source-destination paths taken by packets
- routing algorithms

Circuit Switching

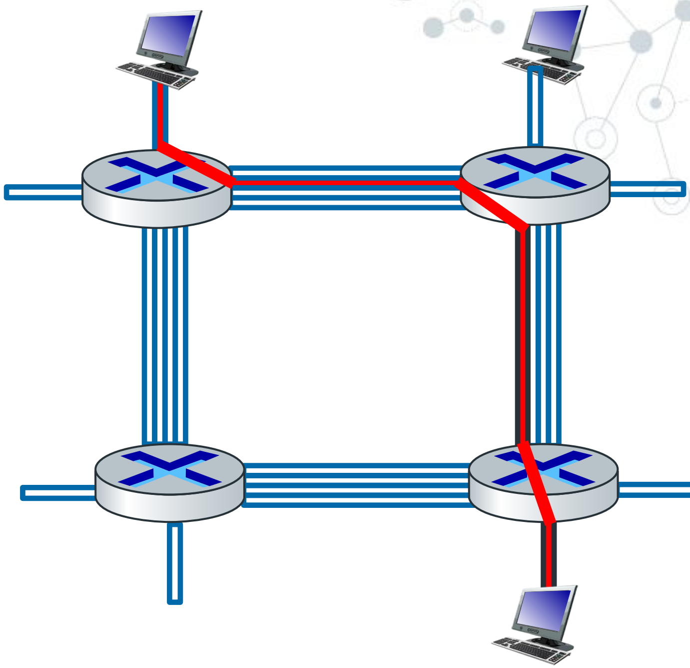
- In circuit-switched networks, the resources needed along a path (buffers, link transmission rate) to provide for communication between the end systems are reserved for the duration of the communication session between the end systems.

- Example:
  - Traditional telephone networks

Circuit Switching

end-end resources allocated to, reserved for “call” between source and destination

- in diagram, each link has four circuits.
  - call gets 2nd circuit in top link and 1st circuit in right link.
- dedicated resources: no sharing
  - circuit-like (guaranteed) performance
- circuit segment idle if not used by call (no sharing)
- commonly used in traditional telephone networks
- if each link between adjacent switches has a



Circuit Switching

circuit switching

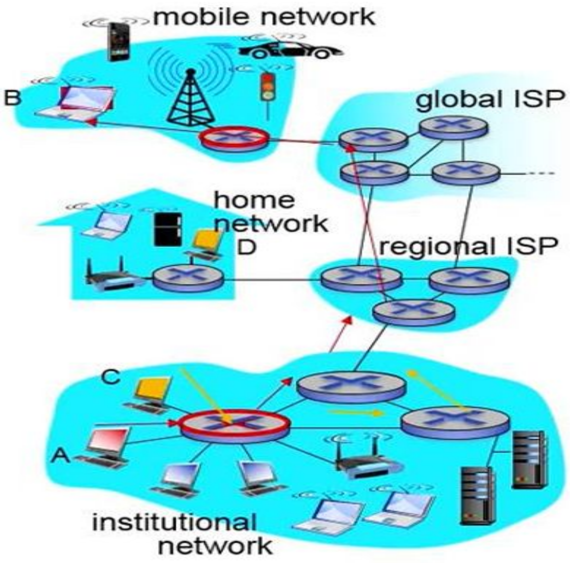
end-end resources allocated to, reserved for "call" between source & dest:

- in diagram, each link 2MB
  - A wants to communicate with B, and needs 1.5MB data rate
    - Call setup
    - Transmit data
  - C wants to D, need 1.5MB
    - Callsetup
    - No callsetup
- dedicated resources: no sharing
  - circuit-like (guaranteed) performance

circuit segment idle if not used by call *(no sharing)*

commonly used in traditional telephone networks

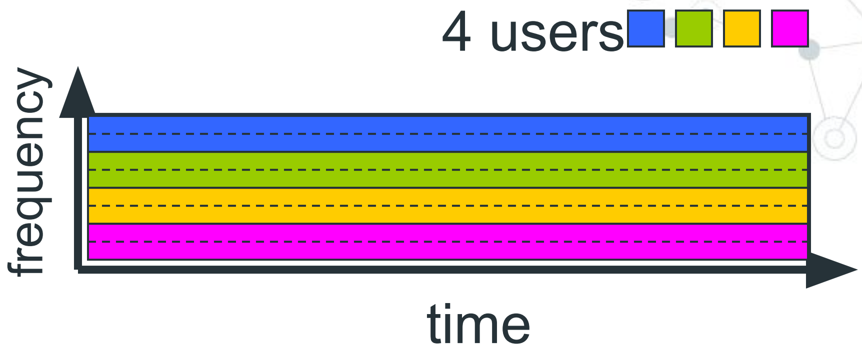
For continuous nature communication



Circuit switching: FDM and TDM

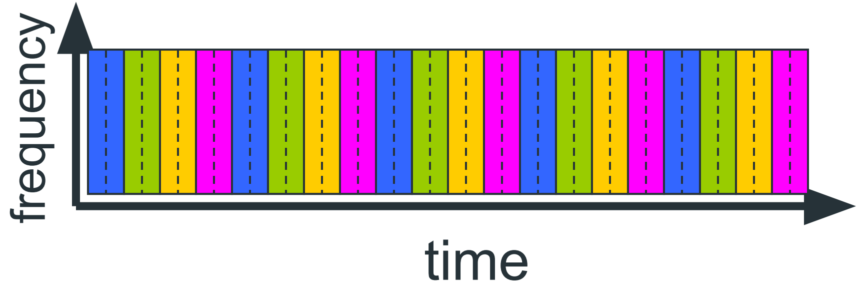
Frequency Division Multiplexing (FDM)

- optical, electromagnetic frequencies divided into (narrow) frequency bands
- each call allocated its own band, can transmit at max rate of that narrow band



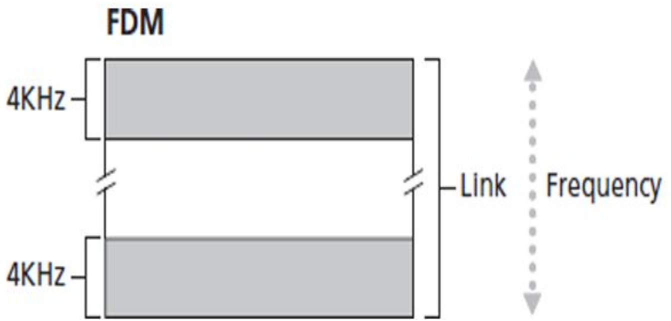
Time Division Multiplexing (TDM)

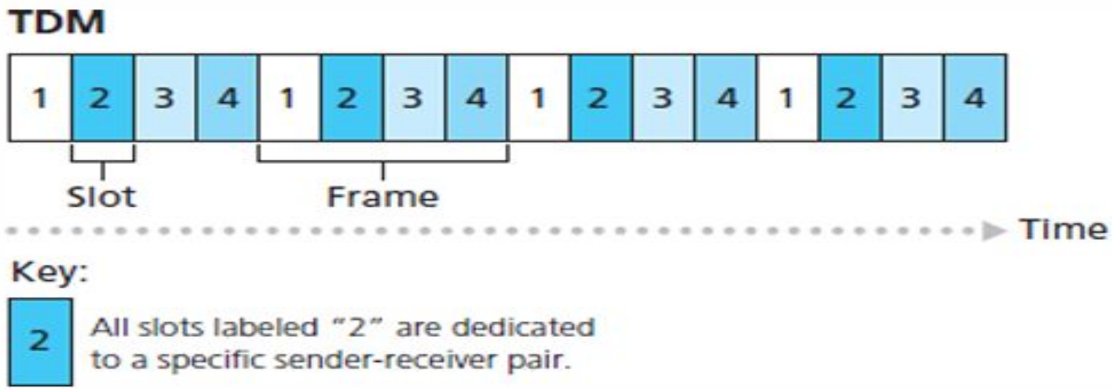
- time is divided into frames of fixed duration
- each frame is divided into a fixed number of time slots
- each call allocated periodic slot(s), can transmit at maximum rate of (wider) frequency band, but only during its time slot(s)



Circuit switching: FDM and TDM

- In telephone networks, this frequency band typically has a width of 4 kHz





Circuit switching: FDM and TDM

For TDM, the transmission rate of a circuit is equal to the frame rate multiplied by the number of bits in a slot.

For example, if the link transmits 8,000 frames per second and each slot consists of 8 bits, then the transmission rate of each circuit is 64 kbps.

Circuit switching: FDM and TDM

Example:

Let us consider how long it takes to send a file of 640,000 bits from Host A to Host B over a circuit switched network. Suppose that all links in the network use TDM with 24 slots and have a bit rate of 1.536 Mbps. Also suppose that it takes 500 msec to establish an end-to-end circuit before Host A can begin to transmit the file. How long does it take to send the file?

Solution:

- Each circuit has a transmission rate of:

  1.536 Mbps/24=64 kbps

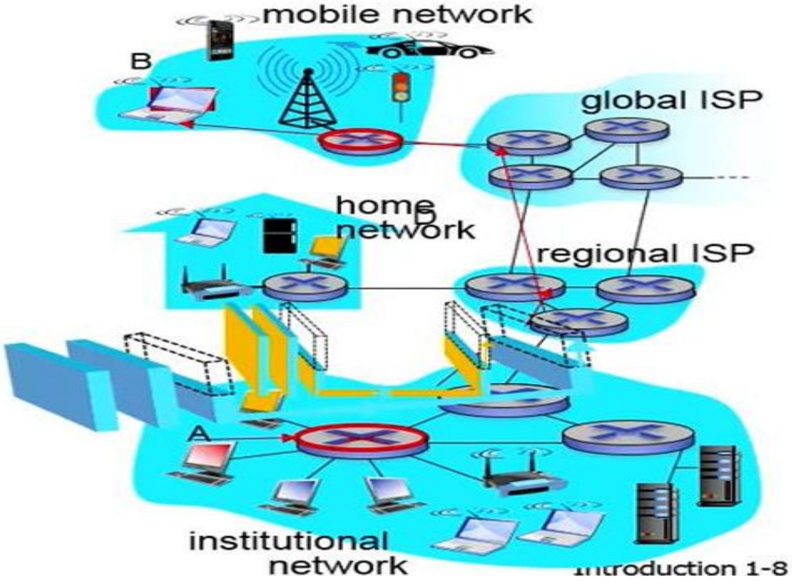
so it takes (640,000 bits)/(64 kbps)=10 seconds to transmit the file. To this 10 seconds we add the circuit establishment time, giving 10.5 seconds to send the file.

Packet switching: store and forward

Packet-switching: store-and-forward

- in diagram, each link 2MB
  - A wants to communicate with B, and needs 1.5MB data rate
    - Transmit data
  - C wants to D, need 1.5MB
- Sharing resources
  - No guaranteed delivery

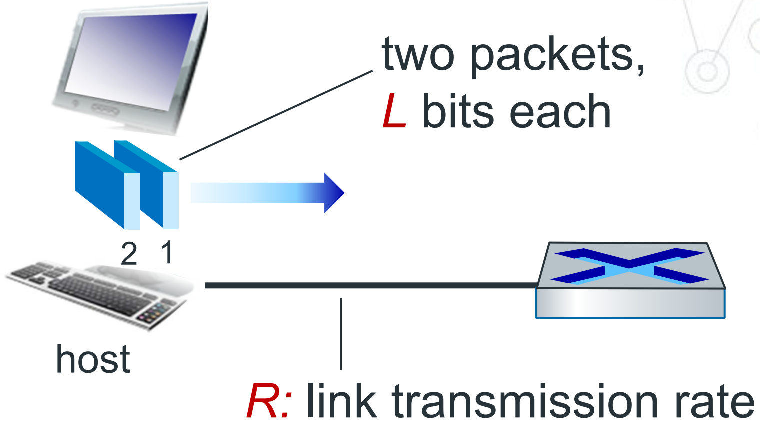
segment idle can b used by others
commonly used in Internet
For bursty data



Host: sends *packets* of data

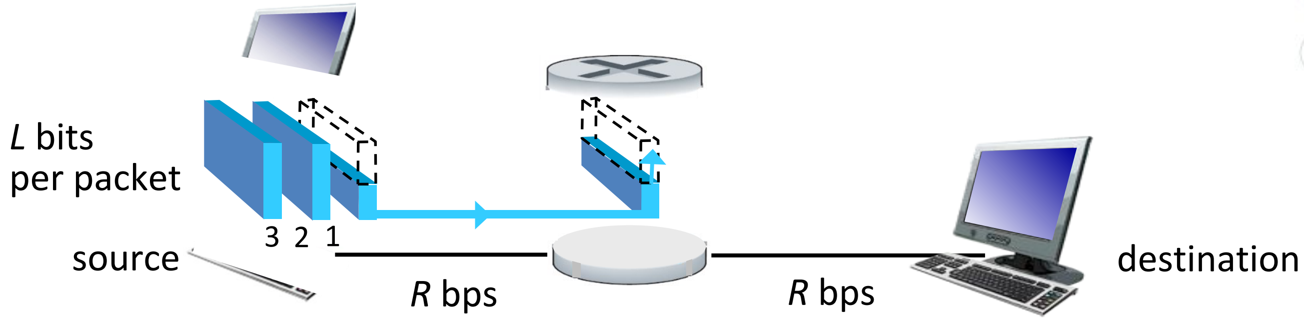
host sending function:

- takes application message
- breaks into smaller chunks, known as *packets*, of length *L* bits
- transmits packet into access network at *transmission rate R*
  - link transmission rate, aka link *capacity, aka link bandwidth*



$$\text{packet transmission delay} = \text{time needed to transmit } L\text{-bit packet into link} = \frac{L \text{ (bits)}}{R \text{ (bits/sec)}}$$

Packet-switching: store-and-forward

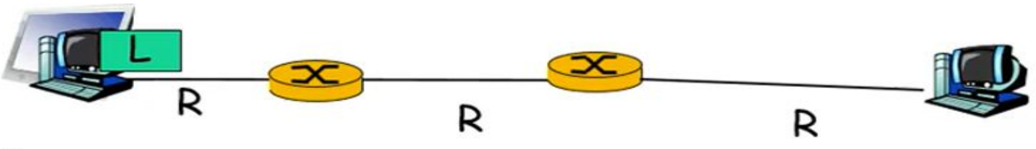


- Transmission delay: takes *L*/*R* seconds to transmit (push out) *L*-bit packet into link at *R* bps
- *Store and forward: entire* packet must arrive at router before it can be transmitted on next link

One-hop numerical example:

- *L* = 10 Kbits
- *R* = 100 Mbps
- one-hop transmission delay = 0.1 msec

Packet switching: store and forward



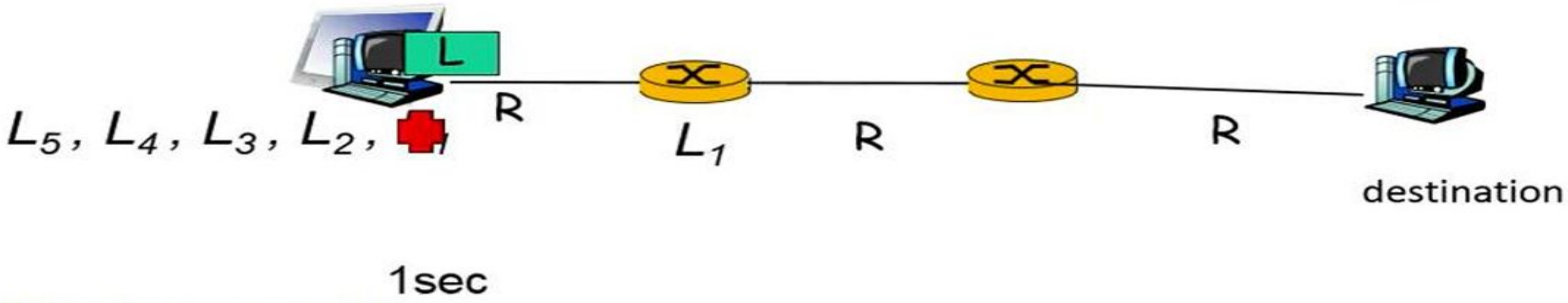
- divides into packet
- Each packet has full address
- takes L/R seconds to transmit (push out) L-bit packet into link at R bps
- *store and forward:* entire packet must arrive at router before it can be transmitted on next link
  - end-end delay = $3L/R$ (assuming zero propagation delay)

more on delay shortly …

one-hop numerical example:

- L = 7.5 Mbits
- R = 1.5 Mbps
- one-hop transmission delay = L/R= 5 sec
- End-to-end delay= 3L/R= 3*5= 15 seconds

Packet switching: store and forward



5 packets example:

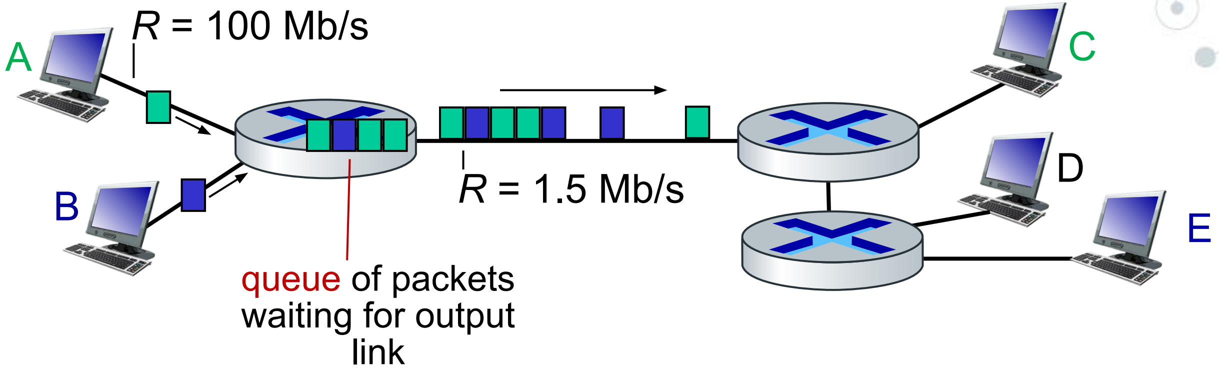
- L = 7.5 Mbits
- Divide it in 5 packets, L_1, L_2, L_3, L_4, L_5,=7.5/5=1.5 Mbits
- 3 hops delay = 7 sec

One packet example:

- L = 7.5 Mbits
- R = 1.5 Mbps
- one-hop transmission delay = 5 sec
- 3 hops delay = 3*5

more on delay shortly …

Packet-switching: queueing delay, loss



Packet queuing and loss: if arrival rate (in bps) to link exceeds transmission rate (bps) of link for a period of time:

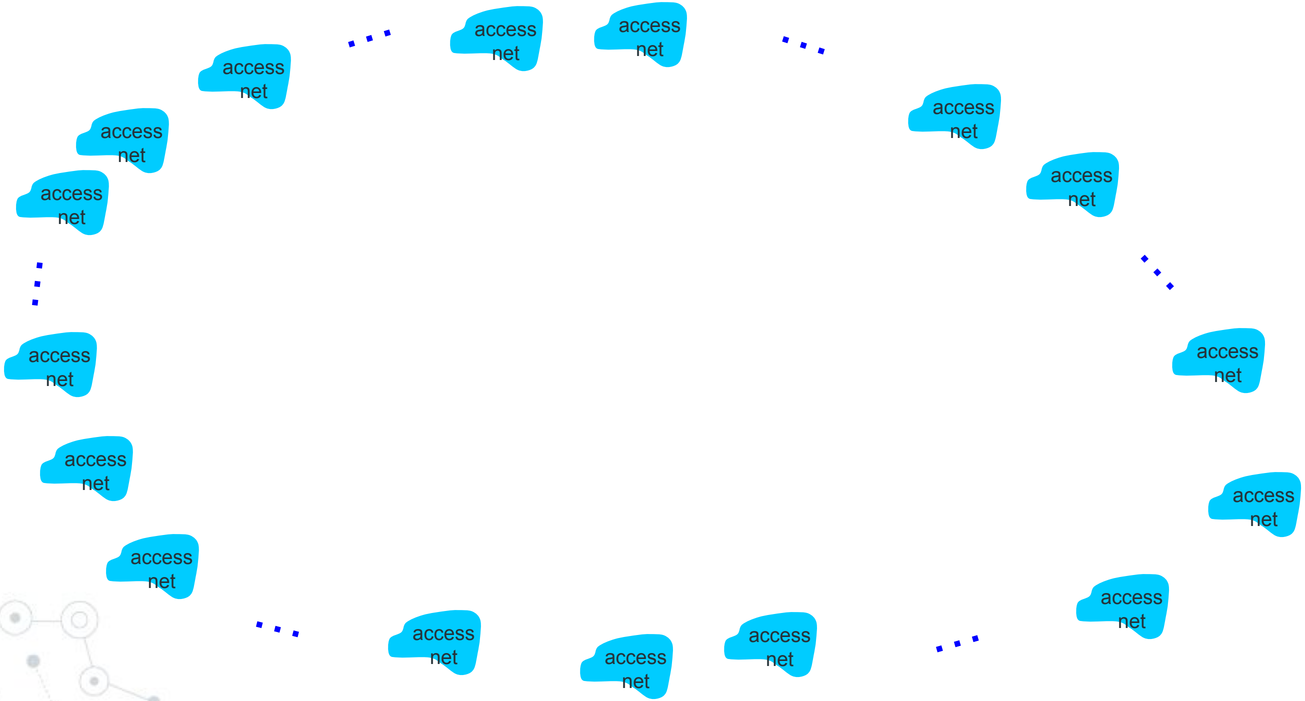
- packets will queue, waiting to be transmitted on output link
- packets can be dropped (lost) if memory (buffer) in router fills up

Internet structure: a “network of networks”

- Hosts connect to Internet via access Internet Service Providers (ISPs)
  - residential, enterprise (company, university, commercial) ISPs
- Access ISPs in turn must be interconnected
  - so that any two hosts can send packets to each other
- Resulting network of networks is very complex
  - evolution was driven by economics and national policies
- Let’s take a stepwise approach to describe current Internet structure

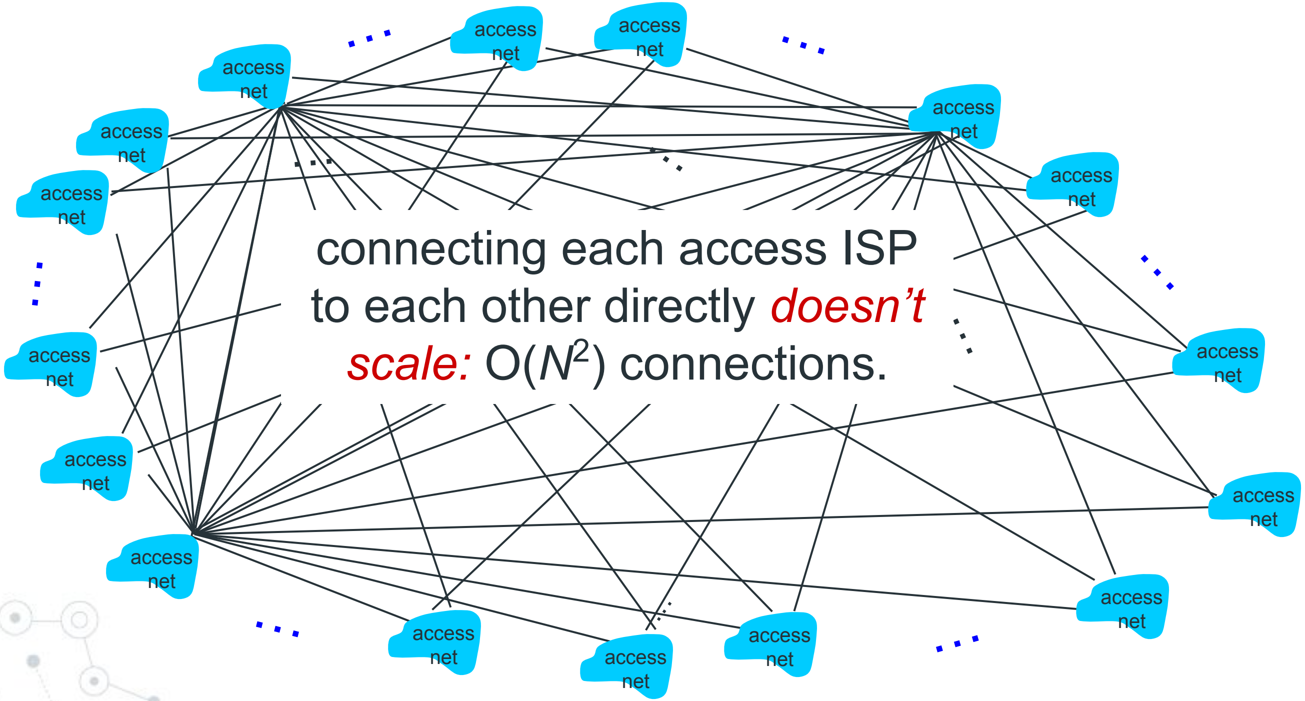
Internet structure: a "network of networks"

Question: given *millions* of access ISPs, how to connect them together?



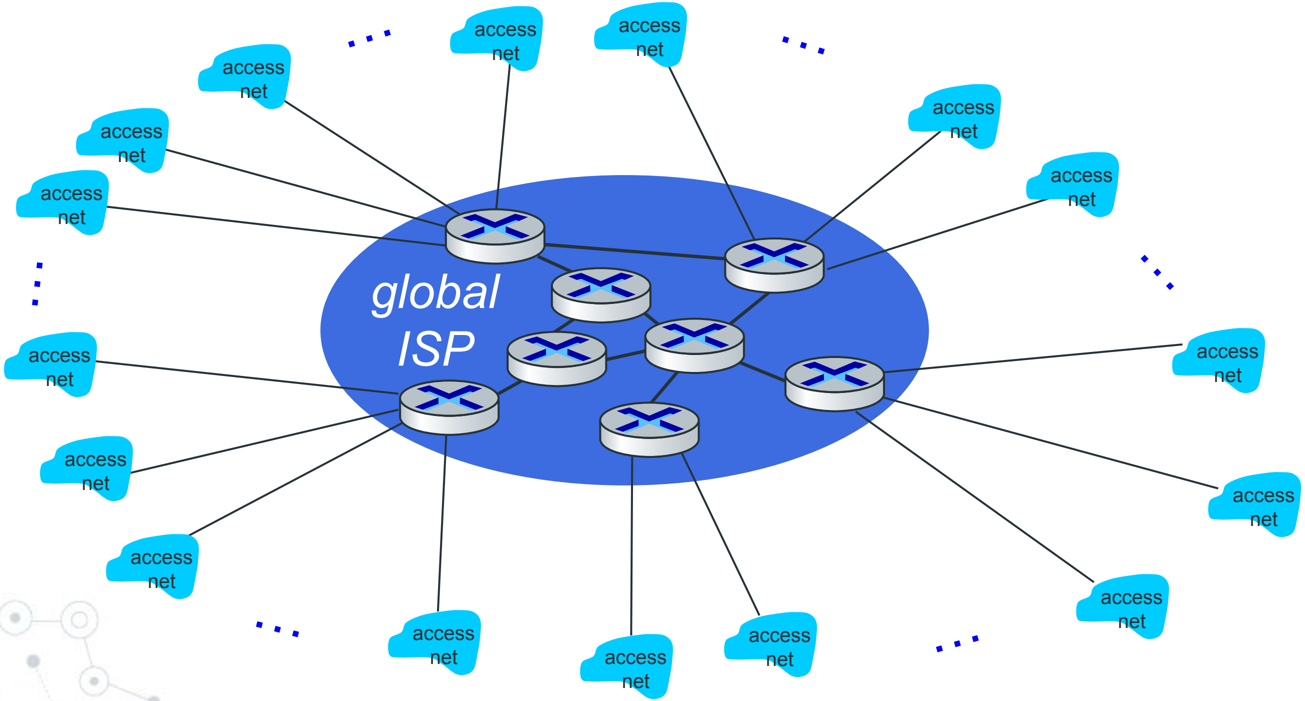
Internet structure: a "network of networks"

Question: given *millions* of access ISPs, how to connect them together?



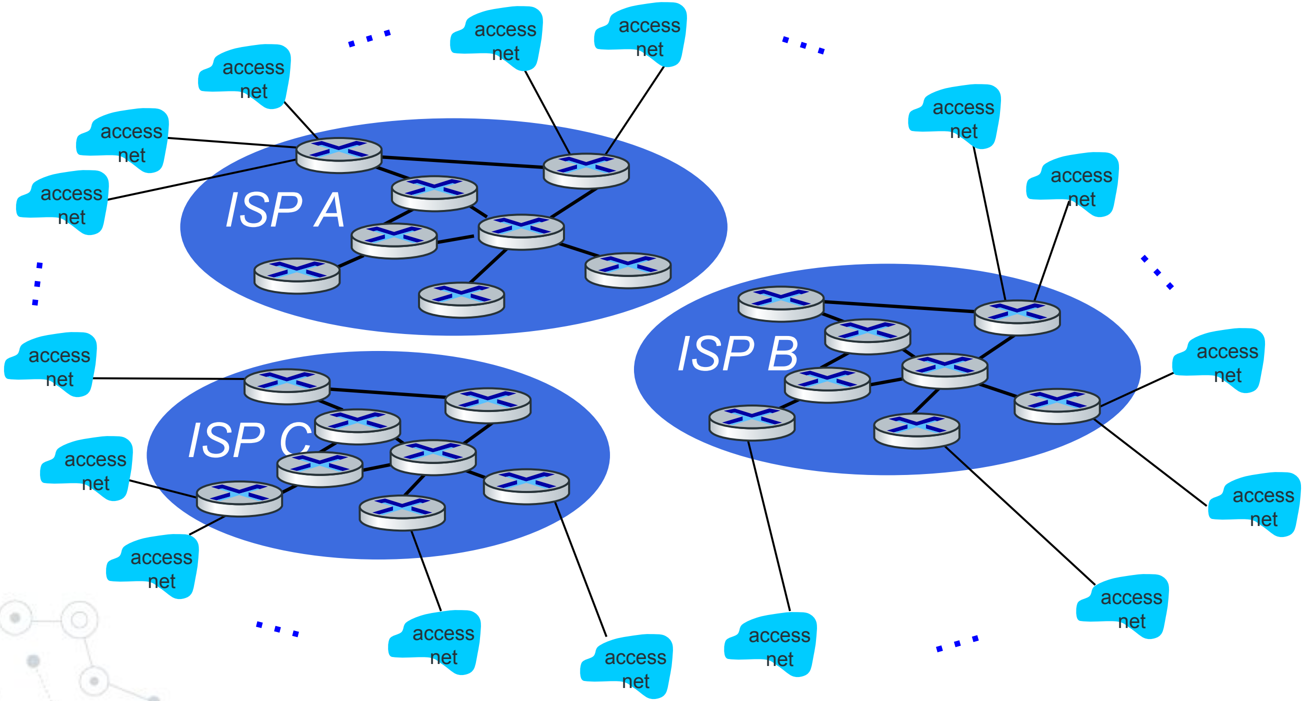
Internet structure: a “network of networks”

Option: connect each access ISP to one global transit ISP?
Customer and provider ISPs have economic agreement.



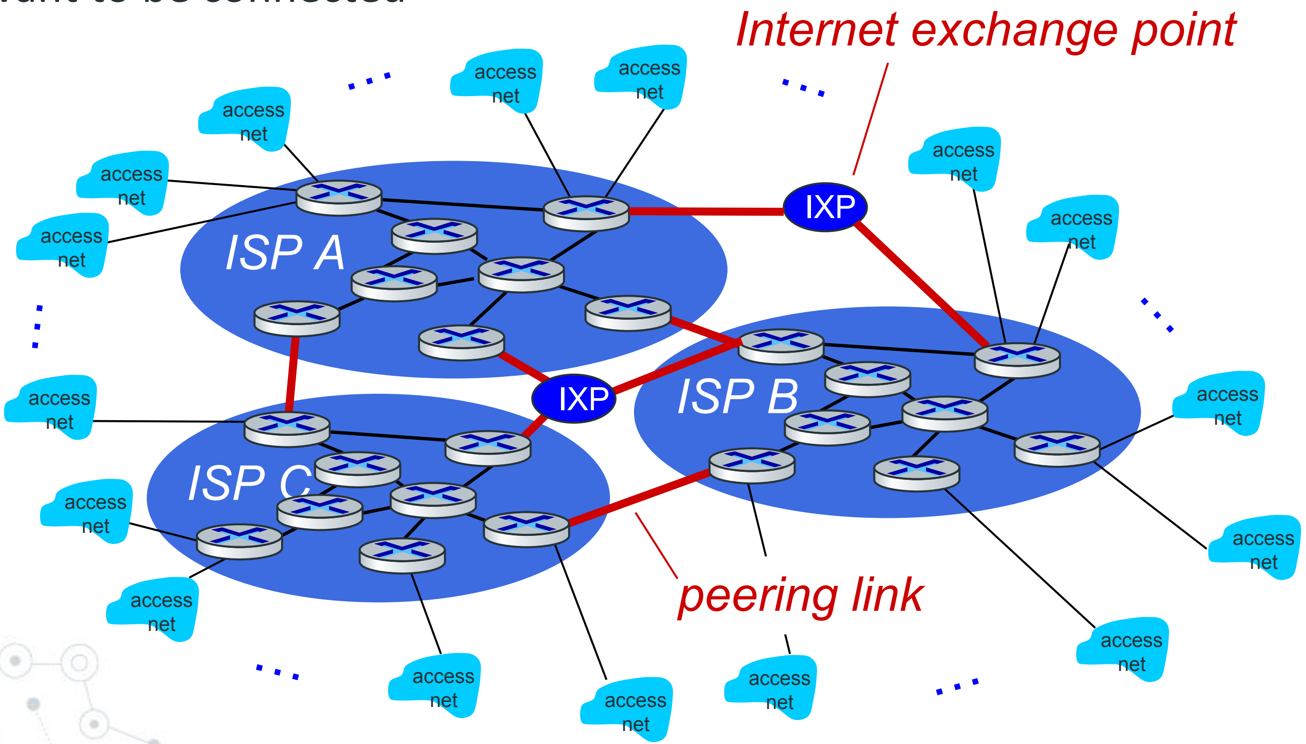
Internet structure: a "network of networks"

But if one global ISP is viable business, there will be competitors ….



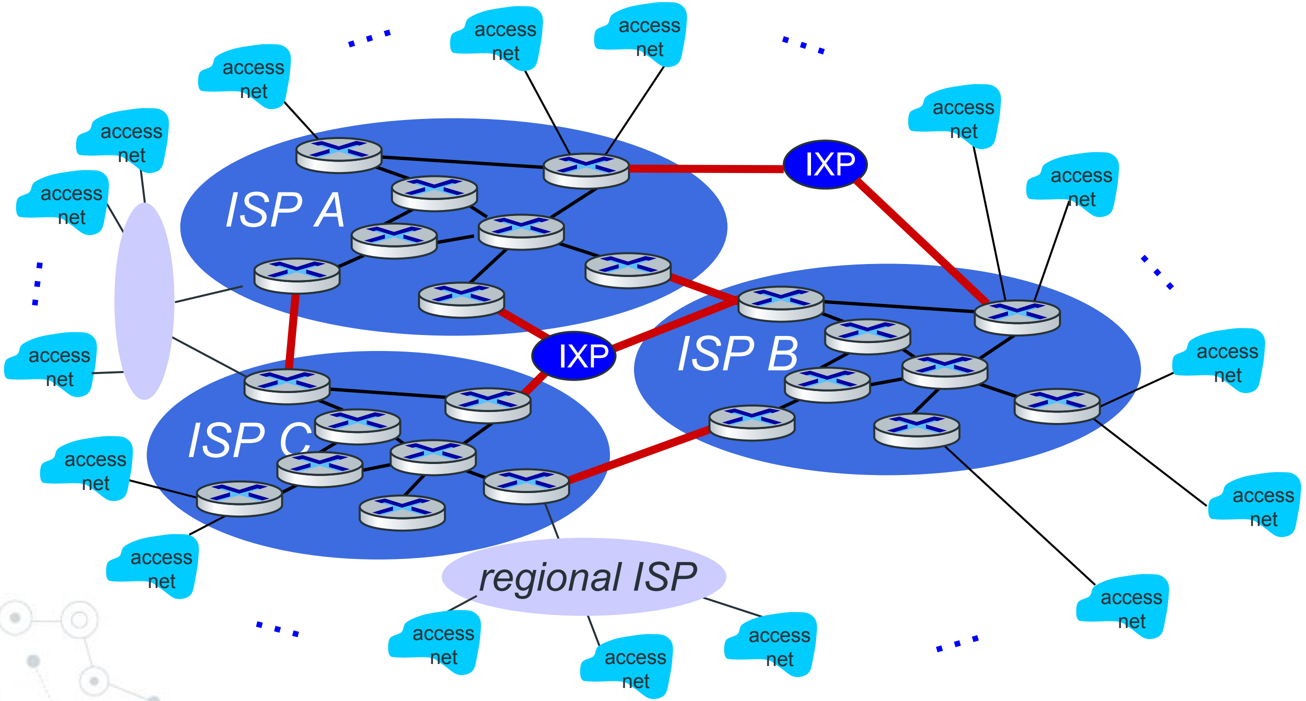
Internet structure: a “network of networks”

But if one global ISP is viable business, there will be competitors …. who will want to be connected



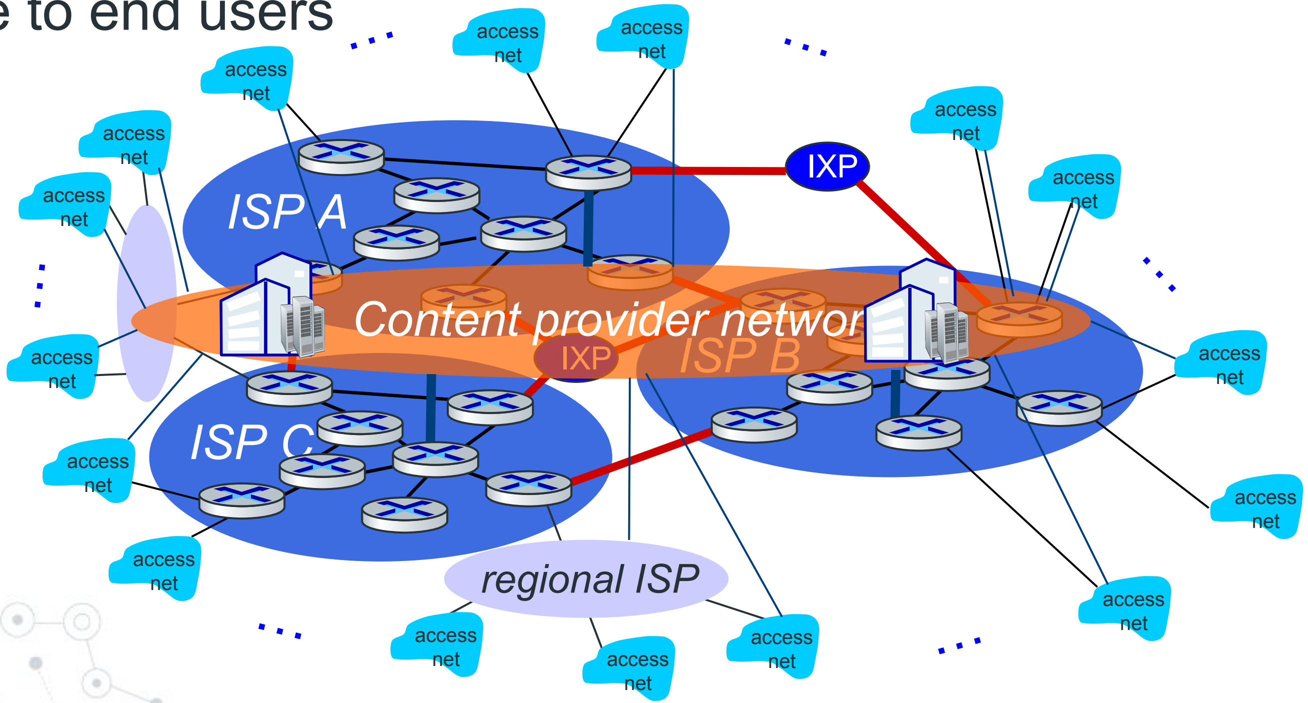
Internet structure: a "network of networks"

… and regional networks may arise to connect access nets to ISPs

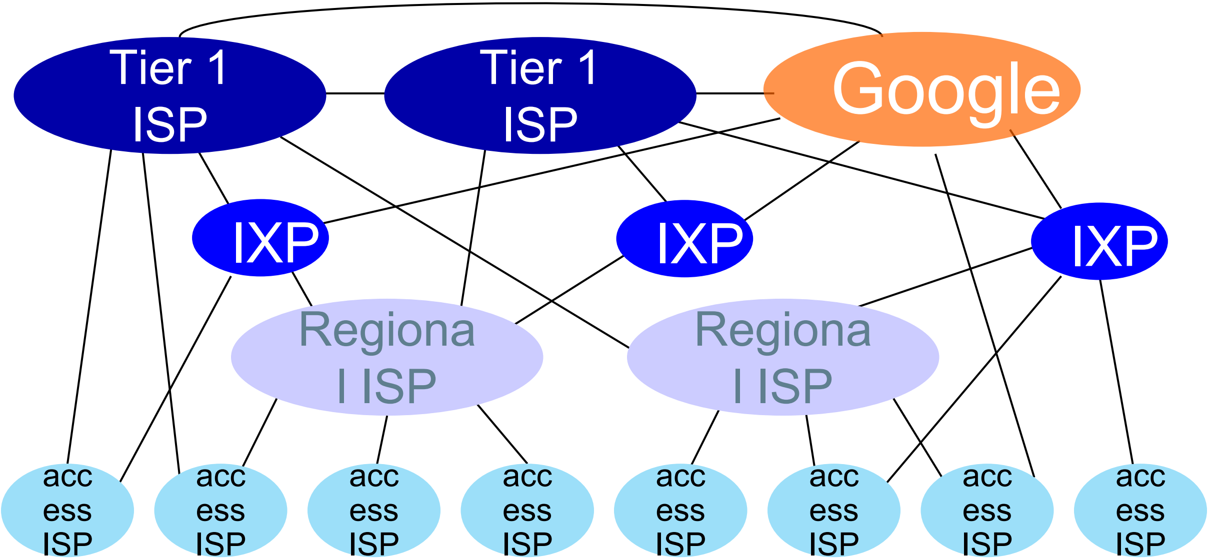


Internet structure: a "network of networks"

… and content provider networks (e.g., Google, Microsoft, Akamai) may run their own network, to bring services, content close to end users



Internet structure: a "network of networks"



At "center": small # of well-connected large networks

- "tier-1" commercial ISPs (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
- content provider networks (e.g., Google, Facebook): private network that connects its data centers to Internet, often bypassing tier-1, regional ISPs

Tier-1 ISP Network map: Sprint (2019)